

# Juan Carlos Tuesta Llaja

## NLP Research Intern — Transformer Interpretability

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### Summary

NLP researcher, recent Master's graduate from MIPT, specializing in interpretability and robustness auditing of Transformer-based language models. Developed AESOP-RU, a syntactically-controlled paraphrase framework (ru-BART-large under SynTagRus constraints), used to audit seven frozen classifiers on Russian text. First-author paper accepted (oral) at Neuroinformatics-2026 (Springer). Fluent in Spanish, Russian, English and Portuguese.

### Education

- 2024–2026 **M.Sc. in Applied Mathematics and Computer Science / Informatics and Computer Engineering**, *Moscow Institute of Physics and Technology (MIPT)*  
Group M05-414B; diploma issued July 2026
- 2019–2024 **B.Sc. in Computer Science (English-track, Phystech School of Applied Mathematics and Informatics)**, *Moscow Institute of Physics and Technology (MIPT)*

### Research Experience

- Oct 2024–Jun 2026 **Graduate Researcher — AESOP-RU: Syntactic Paraphrase Auditing Framework**, *MIPT / ФИЦ ИУ РАН (FRC CSC RAS)*, Supervisors: N.V. Chudova, D.A. Devyatkin
- Designed and trained AESOP-RU, a ru-BART-large model fine-tuned with a custom masked-loss objective enforcing SynTagRus dependency-grammar constraints, for controlled syntactic paraphrase generation in Russian.
  - Built a dependency-to-phrase-structure pipeline (SpaCy, Stanza, SynTagRus) and a paraphrase dataset of ~815,000 pairs (Leipzig Corpora Collection); audited seven frozen Transformer classifiers (RuBERT, mBERT, DistilmBERT, RoBERTa, ruGPT-3, GPT-2, OPT) under controlled syntactic perturbation.
  - Found bidirectional encoders remain stable under heavy syntactic disruption, while causal decoders are markedly more sensitive — syntax causally influences classifier decisions in an architecture-dependent way.
  - Addressed all seven reviewer comments through peer review, resulting in an oral presentation slot at Neuroinformatics-2026.

### Publications

- 2026 **AESOP-RU: Syntactically-Controlled Paraphrase Generation for Behavioral Auditing of Russian-Language Transformer Classifiers**. Tuesta Llaja, J.C., Chudova, N.V., Devyatkin, D.A. *Studies in Computational Intelligence*, Springer. Oral presentation, Neuroinformatics-2026 (BAK Level-1).
- 2026 **Behavioral auditing of Russian-language Transformer classifiers under syntactic masking**. Tuesta Llaja, J.C., et al. *ИТиВС РАН*, Issue 4 (BAK Level-3).

### Technical Skills

- ML / NLP Transformer architectures (BART, BERT, GPT-family), fine-tuning, model interpretability and auditing, dependency/syntactic parsing
- Languages & Tools Python, PyTorch, HuggingFace Transformers, SpaCy, Stanza, Docker, asyncio, Playwright, Git
- AI tools Regular use of LLM-assisted coding and research workflows (Claude, GPT-family) for experiment scripting, literature review, and technical writing
- Languages Spanish (native), Russian (fluent, B2–C1), English (fluent), Portuguese (fluent)